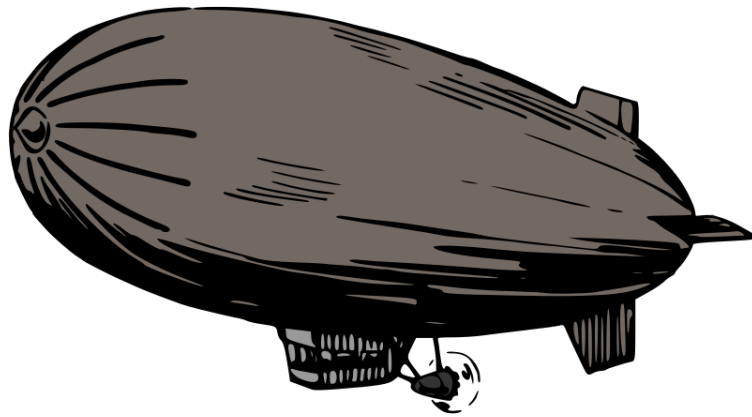


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
SUMMER 2025**



**PROJECT AIRSHIP
AUTONOMOUS CARGO AIRSHIP**

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REVISION HISTORY

Revision	Date	Author(s)	Description
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1 PROBLEM STATEMENT

Today, global logistics and the transportation industry create substantial environmental impact and affect the world in which we live today. With trucks and air transport being the most dominant form of transport for cargo and freight, large amounts of greenhouse gases are emitted every year. The reliance on these forms of transportation on fossil fuel has become a crutch with which we destroy the environment around us to fuel the need for global transportation.

With these facts in mind, it becomes clear that there is a market gap for efficient, cost-effective and environmentally friendly modes of transportation. A mode of transportation that would reduce the reliance on gas and fossil fuels and overcome the unique problems that trucks and planes face.

It would be an even greater success if such a solution could be autonomous and not require human intervention to operate. This would significantly reduce the costs associated with having a human pilot or driver and increase the safety and reliability of this method of transport.

2 METHODOLOGY

For our senior design project, we're building an autonomous cargo airship that delivers freight cost effectively, with low emissions, and independent of infrastructure. It will use helium for lift and electric propulsion, which reduces environmental impact compared to diesel trucks or jet aircraft. The system will have autonomous navigation to plan routes, avoid obstacles, and land safely without human intervention, enabling 24/7 operation and cutting labor costs. Since it doesn't need runways or roads, it can access remote or disaster areas where traditional logistics struggle. Our goal is to create a prototype demonstrating stable flight, cargo handling, and autonomous routing, proving a scalable, sustainable logistics platform that reaches places trucks and planes can't, while lowering costs and emissions.

3 VALUE PROPOSITION

The airship is an excellent project that will model how a real industrial cargo blimp or airship will function. Our model airship will contain many functions that the large-scale airship will use. Our project will not only show that it is possible to create an autonomous airship that can automatically fly, navigate, and land, but that it can detect obstacles and perfectly outmaneuver to get cargo from one destination to another. Our project will tackle problems that a real airship might encounter and demonstrate the benefits and functionality of such a mode of transportation for goods.

Many companies around the world are racing to make a cargo airship such as we are doing, and by funding this project, stakeholders are unlocking new market potential in the transportation industry. All the while supporting a greener alternative to traditional fossil fuel based transportation. This is by all means having a stake in the future of the cargo industry and the world as they move towards environment conscious solutions.

4 DEVELOPMENT MILESTONES

The following is a list of milestones and completion dates:

- Project Charter First Draft - June 2025
- System Requirements Specification - Summer 2025
- Architectural Design Specification - Summer 2025
- Demonstration of Operational Software and Hardware Components - July 2025
- Demonstration of Communication between Software and Hardware - July 2025
- Demonstration of Blimp's Navigation using GPS, Computer Vision, and Sensors - August 2025
- Detailed Design Specification - September 2025
- Demonstration of Airship's Ballonet and Envelope Buoyancy - September 2025
- Demonstration of Airship's Chassis Structure - October 2025
- Demonstration of Complete Implementation of Blimp Components - November 2025
- Demonstration of Autonomous Blimp Performance - November 2025
- Final Project Demonstration - December 2025

5 BACKGROUND

Today's freight transport systems, especially long-haul trucking and air cargo, are struggling with major environmental and operational issues. Heavy-duty trucks and jet-fueled aircraft account for a large portion of logistics-related greenhouse gas emissions, which is speeding up climate change. On top of that, fuel prices are constantly changing, making shipping both costly and unpredictable. Road-based shipping suffers from congestion and infrastructure wear, while air cargo relies on airports and skilled pilots, driving up both costs and complexity especially in remote areas.

These challenges highlight a clear opportunity, the need for a low-emission, fuel-stable, and autonomous logistics system that can move cargo efficiently, even in hard-to-reach areas.

Our proposed autonomous cargo airship aims to fill these gaps. By using helium for lift and electric propulsion for movement, it provides a low-emission alternative to trucks and planes. Electric or hybrid power reduces the risk of exposure to fuel price volatility, while autonomy cuts labor costs and enables 24/7 operation. With minimal ground infrastructure needed, the airship can bypass congested roads and reach places that are tough for trucks or planes to serve such as remote villages, islands, or disaster zones.

This makes the system ideal for a range of users:

- Logistics companies looking to reduce costs and carbon footprints
- Humanitarian organizations needing delivery access to remote regions
- E-commerce platforms aiming to expand long-range delivery options

6 RELATED WORK

Over the past few years, various solutions have been developed to tackle the environmental and logistical issues in global freight transportation using autonomous or eco-friendly aerial systems. Examples of these include:

Hybrid Air Vehicles Airlander 10: Designed to cut emissions, the Airlander 10 commercial hybrid airship uses a mix of aerodynamic and buoyant lift. It can carry up to 10 tons of cargo and has some autonomous features, though it still needs a human pilot. While it shows that low-emission airship transport is feasible, its large size, high cost, and limited autonomy hold back its potential for smaller logistics uses. [4]

Zipline Medical Drone Delivery: Zipline uses self flying drones to deliver medical supplies in hard to reach areas, setting up long range delivery networks that work. Although not airships, Ziplines approach shows that autonomous aerial logistics can be both scalable and reliable. Their system provides useful insights into autonomous flight operations and last mile delivery logistics. [2]

7 SYSTEM OVERVIEW

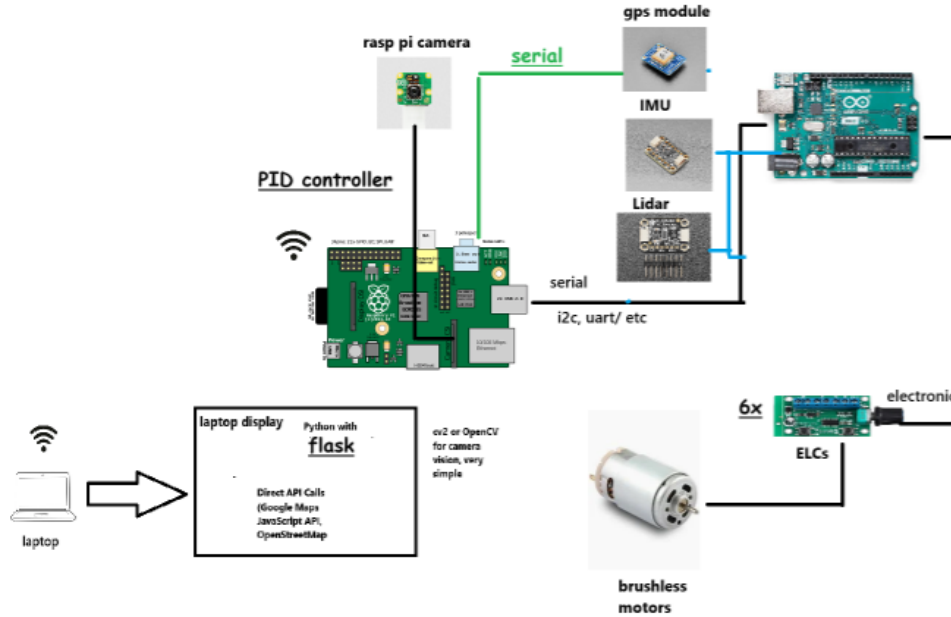
We will make a small model of an autonomous cargo blimp that will one day be manufactured into a large version for companies around the globe. The blimp will be guided using a GPS that will follow a set path. The blimp will automatically lift up with helium and propellers to a set altitude and will traverse from a starting point to its destination. The airship will follow guided way points and detect and avoid obstacles while traveling. In case of emergency, such as a loss of communication signals or power failure, the airship will execute emergency procedures to safely land. [5]

The blimp will consist a rigid frame around a envelope of helium gas and be lifted up with motors attached on the sides. Lightweight and impermeable materials will be used to prevent gas leakage. Landing struts will be connected to the body of the ship to help it safely land. The electrical control systems will sit in the gondola of the airship and will receive continuous telemetry data from multiple sensors. Video data will also be captured and sent to a ground station. Camera vision will be utilized to detect clear landing zones.

Two forward facing propellers will move the airship forward and each individual propeller will control the direction of rotation of the airship. Autonomous logic will be implemented by the ground station and communicated with the airship to allow it to navigate and traverse smoothly through the air.

Automated leveling controllers will ensure that the blimp is leveled at a certain altitude from the ground. The propellers on the airship will speed up and slow down to maintain yaw, pitch, roll, and altitude. [3] Manual controls will be provided to a remote pilot if needed during flight to correct any issues with the flight. Manual controls will individually set the levels for the speed of each propeller and controls over the emergency stop

Power will be managed with enough capacity to power all components for the desired flight time. Power distribution will be necessary and voltage regulators will be used to ensure electrical components receive the correct power they need. A large battery or multiple smaller batteries may be used to power the propellers and the electrical components on board. Battery level will be tracked to ensure when power is low, emergency landing protocols will be activated.



8 ROLES & RESPONSIBILITIES

The stakeholders for this project are us. Our group has a vested interest in the success of this project, even though it may only be a small model of large-scale versions of an autonomous blimp. Our experience in technology and techniques needed to create this autonomous airship (blimp) will be invaluable to many companies seeking to making headway in the transportation industry. Our experience will also be valuable in related fields such as autonomous ground and air transportation, integrating AI and computers to create autonomous solutions.

Arshdeep Singh will be the scrum master and the point of contact for customers. He will be tasked with organizing team meetings and creating a centralized vision for the project. He will also pitch the project to potential customers to communicate the value and the innovation of this project. He will also act as the product owner, defining the specifications for the product and defining how each specification aligns with both business goals and customer needs.

Our team members are:

Team Member	Roles and Responsibilities
Lauren Allbright	Motors Drivers and Aduino controls - Airship Physical Design
Arshdeep Singh	Network Communication - Back up controls - Airship Physical Design
Christian McClain	Sensor Drivers and filtering on Arduino - Airship Physical Design
Brandon Garibay	Server side communication - GUI Controls/Interface
Ahmed Omar	Camera vision and object detection - Control Logic

Other than the basic responsibilities explained above. Team members will do other tasks to make sure this project is a success. In addition to motor drivers and controls, Lauren Allbright will manage the technical needs for the group, making sure every technology used will meet the needs for the project. Christian McClain will simulate the physics to make sure the airship is able to perform the specified aerial actions it requires. Brandon Garibay will maintain the network and data needs for the product and

Ahmed Omar will work with Bradon to make sure that the Ground station controlling the airship keeps control of the aerial vehicle. Arshdeep Singh will design the connections between all the components of the product to meet customer requirements.

The team will maintain the scrum master and the product owner for the whole project. The responsibilities of each member will change and expand throughout the project to create flexibility as each task is assigned to the members most familiar with it.

9 COST PROPOSAL

The approximate budget for the project will be around a \$1050. The majority of the cost will come from a few major components and the cost of materials needed to build the airship. The largest expense will come from the helium cans that will contain the gas needed to make the airship lift. Since a considerable amount of gas will be needed to build the product, multiple helium gas canisters will be needed and stored. Other expensive components will include the carbon fiber rods needed to construct the frame of the airship and the electronic components such as sensors, micro-controllers, batteries, and motors.

9.1 PRELIMINARY BUDGET

Below is a table listing components and their costs needed to create the airship. Each item has a fixed cost and duplicate items' cost has been combined into one line item. We expect not all of the budget of the project to be used for the building materials.

Item	Cost
Raspberry Pi	\$64.99
Arduino Uno	\$27
Raspberry Pi Camera Module 3	\$29.25
Hot Glue Sticks	\$1.46
Xylene solvent	\$10.99
Mylar Emergency Blanket 4 pack	\$3.99
GPS Hat for Raspberry Pi	\$29.95
IMU 20948	\$14.95
Adafruit VL53L0X Time of Flight Distance	\$14.95
Adafruit BMP388	\$9.95
3x STEMMA QT 4-pin cable	\$2.85
QwinOut motor propellers	\$12.38
QwinOut 2-4S 30A RC Brushless ESC	\$37.98
QwinOut A2212 1000KV Motors	\$31.85
OVONIC 2s Lipo Battery 50C 2200mAh	\$16.92
5xCarbon Fiber Rods	\$34.95
Total	\$349.96

9.2 CURRENT & PENDING SUPPORT

There are two sources of funding for this project. The funding sources are listed below and the approximate amount from each source of funding. All funding sources has been secured and there are no pending funding sources for this project.

Source	Amount
UTA Senior Design Project budget	\$800
5 x Group member personal budget	\$250
Total	\$1050

10 FACILITIES & EQUIPMENT

The facilities we will be using are:

- Lab Maker Space
- ERB Labs
- Cricket Pitch Field
- Library
- Senior Design Drone Cage
- SD Lab 203
- University Center

We will utilize these spaces to construct and test our blimp. Other spaces may help us with creating and working on specific components such as the ERB labs which often contain equipment like Voltmeters, Oscilloscopes, and soldering Irons that we may be using. We will fully utilize the UTA maker space to create the physical body of our blimp and may utilize tools and resources in the maker space. We will also be using the UTA library to conduct research, meet with group members and stakeholders, and design documents for our project.

Furthermore, we will utilize places like the SD lab 203 for research on other projects that meet the criteria of autonomous vehicles to get ideas to implement in our project or to build off the techniques used by previous senior design teams in designing our blimp. We will also use this space to store and retrieve our project when working and utilize the lab personnel in getting ideas and resources in helping us understand the challenges we may face in building our project.

To fully test and qualify our blimp, we will use multiple locations during many stages of our project. At the beginning we will utilize public spaces like the University Center to work on electrical components of our project, such as motors, micro-controllers, and sensors. After testing these components we will begin to put these components together along with the physical body of the blimp itself. We will utilize the Senior Design Lab's drone cage to further test the aerial capabilities of our design and make modifications needed to take our blimp in the outside environment. To finally test the full capabilities and to demonstrate the abilities of our blimp, we will utilize the UTA cricket pitch.

The full list of equipment we may use while working on our project is:

- Oscilloscopes
- Soldering Iron
- CNC Machines
- 3D Printer
- Storage Trunk
- Voltmeters

11 ASSUMPTIONS

The following list contains critical assumptions related to the implementation and testing of the project:

- Knowledge and basic implementation of the hardware and software components will be completed by the 3rd sprint cycle.
- Communication and functionality between the aircraft's components will be done by the 4th sprint cycle.
- The airship's ballonnet and envelope will be constructed by the early fall semester.
- The airship's frame, gondola, and chassis will be assembled by the fall semester.
- UTA's Cricket Pitch field and drone cage will be available by the fall semester.

12 CONSTRAINTS

The following list contains key constraints related to the implementation and testing of the project:

- Resources: A limited amount of helium is available for testing and final demonstration.
- Deadline: The autonomous blimp must be fully functional for demonstration by December 3rd, 2025.
- Environment: The weather, the range of the wireless network, and the test area will not always be ideal for flying.
- Budget: Total development costs, such as hardware, software, and aircraft components, must not exceed \$800.
- Compliance: The autonomous airship must follow the Federal Aviation Administration guidelines [1].

13 RISKS

The following high-level risk census contains identified project risks with the highest exposure. Mitigation strategies will be discussed in future planning sessions.

Risk description	Probability	Loss (days)	Exposure (days)
Poor Internet Connection	0.65	1	0.65
Bad Weather during testing	0.40	3	1.2
High Wind Speeds	0.70	1	0.7
Improper Balancing	0.20	5	1
Rip in the blimp	0.65	7	4.55

Table 1: Overview of highest exposure project risks

14 DOCUMENTATION & REPORTING

14.1 MAJOR DOCUMENTATION DELIVERABLES

14.1.1 PROJECT CHARTER

The project charter will outline the scope, goals, timeline, and deliverables of the project. The first version will be completed by the end of Sprint 1, the 25th of June. Updates will be made if major changes occur in the scope of the project. The entire group will contribute to maintaining the document based on each individual's knowledge. The final version will be due early December.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATION

The System Requirements Specification will define all functional and non-functional requirements for the project. The initial version will be completed by the end of Sprint 2, July 9th. The document will be updated if any requirements change or are added throughout the development of the project. The final version will be due early December.

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION

The Architectural Design Specification will describe the project's hardware and software components, interfaces, and data flow. The initial version will be delivered by the end of Sprint 3, July 23rd. The document will be updated if the design needs to be refined or changed throughout development. The final version will be due early December.

14.1.4 DETAILED DESIGN SPECIFICATION

The Detailed Design Specification will expand on the architecture document. Information on individual components, algorithms, and interface details will be provided. The initial version will be completed by the end of Sprint 5, in late September, and will be updated if necessary as the components are implemented. The final version will be due early December.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

The Product Backlog will be created from the SRS, taking each requirement into consideration. The items will be prioritized by group vote, giving more weight to the items necessary to allow every group member to contribute. We will use Teams and Discord to maintain and share the Product Backlog with team members.

14.2.2 SPRINT PLANNING

Sprint planning will begin after the end of the previous sprint. We plan on having 8 sprints throughout the project. The team will review the backlog and decide which items take priority for the next sprint when planning.

14.2.3 SPRINT GOAL

The sprint goal will be set during sprint planning. The team will need to agree on what goal is necessary for the project to move forward. This will also be heavily influenced by Documentation deliverables needs.

14.2.4 SPRINT BACKLOG

Sprint Backlog items will be selected by the team during sprint planning. The backlog will be maintained using Teams and Discord for communication.

14.2.5 TASK BREAKDOWN

Tasks will be self-assigned from the sprint backlog based on interest and expertise. Time spent on tasks will be logged individually on sprint review presentations.

14.2.6 SPRINT BURN DOWN CHARTS

The team member who presents for Sprint Review will be responsible for generating the burn down chart for that sprint. Each individual will be held responsible for reporting the amount of effort expended.

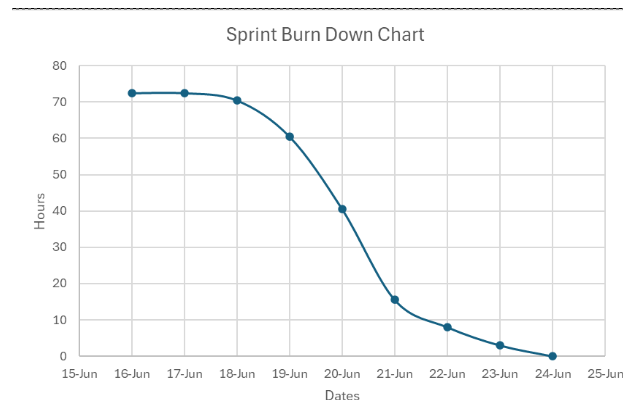


Figure 1: Example sprint burn down chart

14.2.7 SPRINT RETROSPECTIVE

Sprint Retrospectives will be held after the end of each sprint along with Sprint planning. If any major issue or topic is brought up that will be documented in a collective document. Each individual will need to take into account the Sprint Retrospective when completing their own Individual Sprint Report, due two days after each sprint ends.

14.2.8 INDIVIDUAL STATUS REPORTS

Each individual will report the sprint goal and backlog as well as what they worked on in the backlog. An Individual Retrospective will be completed evaluating their contributions. A peer review will be written of each team member, assessing different aspects of each member.

14.3 CLOSEOUT MATERIALS

14.3.1 SYSTEM PROTOTYPE

The final system prototype will include fully integrated hardware and software capable of demonstrating the project's requirements.

14.3.2 PROJECT POSTER

The poster will summarize the project's goals, design, implementation, and results. It will include photos and system diagrams. It will be presented in early December.

14.3.3 WEB PAGE

The web page will include an overview of the project, results and demo videos. It will be accessible to the public and will be delivered at closeout, around December.

14.3.4 DEMO VIDEO

The Demo Video will show all of the blimps functionality and how it is controlled. The video will be a few minutes, just as long as needed to show how the blimp works.

14.3.5 SOURCE CODE

The source code will be maintained using a GitHub Repository under an MIT license, or other licenses required.

14.3.6 SOURCE CODE DOCUMENTATION

Source code documentation will be generated using Doxygen and other resources. A Documentation will be available as PDF and HTML.

14.3.7 HARDWARE SCHEMATICS

We will be wiring components together.

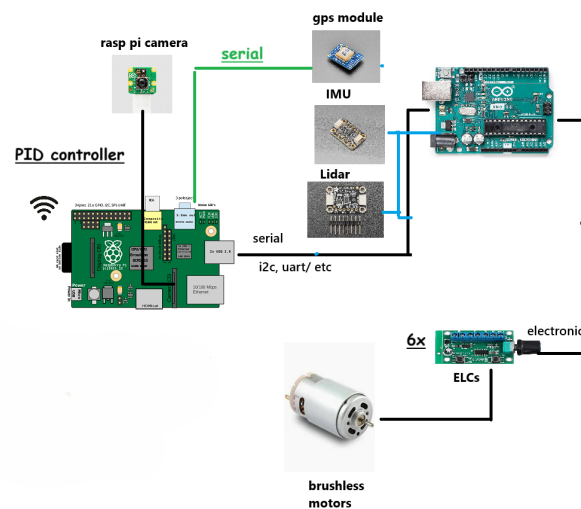


Figure 2: Wiring Diagram

14.3.8 CAD FILES

Mechanical designs for various parts will be created using Tinkercad to generate the 3D files.

14.3.9 INSTALLATION SCRIPTS

Installation instructions will be available in the README file of the GitHub repository. This will guide users on how to set up the blimps software environment and how to run it.

14.3.10 USER MANUAL

The User Manual will be included in the README file in the repository. This will have instructions on how to set up the system environment and how to use the blimp.

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